

README: Reproducibility Instructions for: "Assessing Public Preferences for Reforming Water Subsidy Eligibility Rules in Delhi"

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Overview

This readme contains instructions for reproducing the analysis of the paper, "Assessing Public Preferences for Reforming Water Subsidy Eligibility Rules in Delhi"

The folder structure is as follows:

Folder name	Description of contents
Analysis_R_Code	All the code required for running the multinomial logit and latent class models
Analysis_R_Package	Contains required packages, and all dependencies for archiving
Data	Data required for running the models
Output_Apollo	The output from running all the files in Analysis_R_Code folder
Output_Tables_Fig_Manuscript	These contain the formatted tables and figures produced.
Start_Search_Files_and_Outputs	Files and outputs from the start searches conducted to determine optimal start values and produce the results of the final latent class model. This contains the original .RDS file of the model output.

Operating system requirements:

Any operating system with the ability to install and run R. The original analysis was conducted in Windows 11.

Software and versions:

The original analysis was conducted on R 4.3.2. It was tested on R 4.6.0

The main R package required for replication is Apollo. The original analysis was conducted on Apollo Version 0.3.8.

Apollo package information on Hess & Palma (2019) DOI 10.1016/j.jocm.2019.100170

www.ApolloChoiceModelling.com

Start search operating system, software and versions:

The start search was performed on a university server using R 4.4.0 for Linux, Apollo 0.3.8.

Packages required:

All the packages required for the models (Table 1 and 2), and all their dependencies are in the folder **Analysis_R_Package**

Table 1: Packages required for the main analysis

Package	Version
apollo	0.3.8
openxlsx	4.2.8.1
dplyr	1.2.1

Table 2: Additional packages required for the formatted figures and tables in the manuscript

Package	Version
stringr	1.6.0
ggplot2	4.0.3
ggtext	0.1.2
tidyr	1.3.2
scales	1.4.0
ggrepel	0.9.8
ggh4x	0.3.1
forcats	1.0.1
rstudioapi	0.19.0

Requirement for Non-standard Hardware

There are no unusual computational requirements; a conventional PC/laptop suffices to run the main models in the folder **Analysis_R_Code** and to load the model results from the start search conducted which is in the folder **Start_Search_Files_and_Output**.

The start search, based on which the results in the main paper are presented was run on a university server on June 23, 2025. Running this would require high computing power as it takes a long time to complete. The configurations used are in the folder, in the file SS16_SubElig_Class_3_20260613_1+1_Q1_HH.sh.

Installation Guide

- R can be downloaded from <https://www.r-project.org/> and then installed
- R Studio can be downloaded from <https://posit.co/download/rstudio-desktop/> and must be installed after R is installed.
- The Apollo package version 0.3.8 can be accessed in the Analysis_R_Package folder in this reproducibility package. The latest version would be installed if R is used to install it

directly. Documentation for any subsequent version, can be found at

<https://www.apollochoicemodelling.com/code.html>

- The latest version of all packages can be installed using `install.packages("package_name")`, in which you replace "package_name" with the name of the required packages above.
- Package versions used in the original analysis with dependencies are in the folder **Analysis_R_Package**

The typical install time on a "normal" desktop computer is about 10 minutes total.

Instructions for running code and data

- Open **Analysis_R_Code** and open any of the R studio files
 - `SS_SubsidyEligibility_LCM.r`
 - `SubsidyEligibility_01_MNL.r`
 - `SubsidyEligibility_02_MNL_Version.r`
 - `SubsidyEligibility_03_MNL_SQ.r`
 - `SubsidyEligibility_04_LCM.r`
 - `SubsidyEligibility_05_LCM_Ver.r`
 - `SubsidyEligibility_06_LCM_SQ.r`
 - `SubsidyEligibility_07_LCM_DK.r`
- Install required packages if needed using `install.packages("package_name")`, in which you replace "package_name" with the name of the required packages
- Open and Run (click on "Run" at the top of the R Studio file, or press "Ctrl+A" to select all, and then press "Ctrl+ENTER") each of the files in the folder separately, which will produce Excel and text files as outputs from the models.
- Package versions used in the original analysis with dependencies are in the folder **Analysis_R_Package**
- The expected output from running `SS_SubsidyEligibility_LCM.r` can be found in the **Output_Tables_Fig_Manuscript** folder. This loads the model run after a start search to reproduce the tables and figures in the manuscript. The outputs are:
 - `SS_Table_3.xlsx`
 - `SS_Figure_3_PP.png`
 - `SS_Extended_Figure_1_Log_Odds.png`
- The expected output from running the remaining models can be found in the **Output_Apollo** and **Output_Tables_Fig_Manuscript** folders. These run the models

without loading the output model and without a start search. They may produce numbers that vary by 0.01 decimal places based on the package versions and operating system.

Outputs in the **Output_Apollo** folder:

- SubsidyEligibility_01_MNL_estimates.csv
- SubsidyEligibility_01_MNL_iterations.csv
- SubsidyEligibility_01_MNL_model.rds
- SubsidyEligibility_01_MNL_output.txt
- SubsidyEligibility_02_MNL_Version_estimates.csv
- SubsidyEligibility_02_MNL_Version_iterations.csv
- SubsidyEligibility_02_MNL_Version_model.rds
- SubsidyEligibility_02_MNL_Version_output.txt
- SubsidyEligibility_03_MNL_SQ_estimates.csv
- SubsidyEligibility_03_MNL_SQ_iterations.csv
- SubsidyEligibility_03_MNL_SQ_model.rds
- SubsidyEligibility_03_MNL_SQ_output.txt
- SubsidyEligibility_04_LCM_estimates.csv
- SubsidyEligibility_04_LCM_iterations.csv
- SubsidyEligibility_04_LCM_model.rds
- SubsidyEligibility_04_LCM_output.txt
- SubsidyEligibility_05_LCM_Ver_estimates.csv
- SubsidyEligibility_05_LCM_Ver_iterations.csv
- SubsidyEligibility_05_LCM_Ver_model.rds
- SubsidyEligibility_05_LCM_Ver_output.txt
- SubsidyEligibility_06_LCM_SQ_estimates.csv
- SubsidyEligibility_06_LCM_SQ_iterations.csv
- SubsidyEligibility_06_LCM_SQ_model.rds
- SubsidyEligibility_06_LCM_SQ_output.txt
- SubsidyEligibility_07_LCM_DK_estimates.csv
- SubsidyEligibility_07_LCM_DK_iterations.csv
- SubsidyEligibility_07_LCM_DK_model.rds
- SubsidyEligibility_07_LCM_DK_output.txt

Outputs in the folder **Output_Tables_Fig_Manuscript**:

- SubsidyEligibility_01_MNL_Table_3.xlsx
- SubsidyEligibility_02_MNL_Version_Extended_Table_4.xlsx
- SubsidyEligibility_03_MNL_SQ_Extended_Table_4.xlsx
- SubsidyEligibility_04_LCM.xlsx

- SubsidyEligibility_04_LCM_Log_Odds.png
- SubsidyEligibility_04_LCM_PP.png
- SubsidyEligibility_05_LCM_Ver.xlsx
- SubsidyEligibility_06_LCM_SQ.xlsx
- SubsidyEligibility_07_LCM_DK.xlsx

Description of variables

Variable	Description	Levels	Description of Levels
ID	Household ID		
Weights	Scaled weights to account for unequal probabilities of selection in the clusters		
Informal_Settlement	Indicator showing whether the household was in an informal settlement or planned neighborhood	1 0	Informal settlement Planned neighborhood
hh_size	Number of household members		
hh_size_60	Number of household members 60 years of age or more		
hh_caste	Variable indicating whether the household belonged to a marginalized caste	Yes No Missing	A marginalized caste Not a marginalized caste Missing value
hh_diff_abled	Variable indicating whether the household had differently-abled members.	Yes No Missing	Had differently-abled members Did not have differently-abled members Missing value
Wealth_Quin	Wealth quintiles using a wealth index		1 to 5, corresponding to wealth groups. 1 is the lowest wealth, and 5 is the highest wealth quintile
Got_Subsidy	Variable indicating whether the household received the water subsidy at least once in a year. "DK" was used to indicate that the respondent did not know if they received the subsidy or not.	Yes	Received the water subsidy at least once a year

		No	Did not receive the water subsidy at all
		DK	Respondent did not know if they received the water subsidy or not
BWS_1_Version	Variable indicating whether the respondent received the BWS questions framed in terms of fairness or preference	1	Fairness version
		0	Preference version
BWS_1_SQ	Variable indicating whether the status quo (current policy) was the first option or the last option in the set	1	Status quo (current policy) as the first option
		0	Status quo (current policy) as the last option
BWS_1_Block	Indicating the block number (from 1 to 15) that was assigned to the respondent		
BWS_1_F_Most	Variable indicating which basis for the subsidy was chosen as the First-Best option by the respondent	1	Income
		2	Household size
		3	Caste
		4	Elderly or differently-abled
		5	Not in planned neighborhood
		6	10,000 L/month unconditional subsidy
		7	Current subsidy policy (20,000 L/month for connected households)
BWS_1_F_Least	Variable indicating which basis for the subsidy was chosen as the First-Worst option by the respondent	1	Income
		2	Household size
		3	Caste
		4	Elderly or differently-abled

		Not in planned 5 neighborhood 10,000 L/month 6 unconditional subsidy Current subsidy policy (20,000 L/month for 7 connected households)
BWS_1_S_Most	Variable indicating which basis for the subsidy was chosen as the Second-Best option by the respondent	1 Income 2 Household size 3 Caste 4 Elderly or differently-abled Not in planned 5 neighborhood 10,000 L/month 6 unconditional subsidy Current subsidy policy (20,000 L/month for 7 connected households)
BWS_1_S_Least	Variable indicating which basis for the subsidy was chosen as the Second-Worst option by the respondent	1 Income 2 Household size 3 Caste 4 Elderly or differently-abled Not in planned 5 neighborhood 10,000 L/month 6 unconditional subsidy Current subsidy policy (20,000 L/month for 7 connected households)
Inc	If income-based subsidy was available as an option for the Block	Available as an option in 1 the block Not available as an option 0 in the block
HH	If household size-based subsidy was available as an option for the Block	Available as an option in 1 the block Not available as an option 0 in the block
Caste	If a caste-based subsidy was available as an option for the Block	Available as an option in 1 the block

		0	Not available as an option in the block
ElderDiffAble	If subsidy targeted toward households with elderly or differently abled members was available as an option for the Block	1 0	Available as an option in the block Not available as an option in the block
NotPlanned	If subsidy targeted toward households in informal settlements was available as an option for the Block	1 0	Available as an option in the block Not available as an option in the block
First10k	If the policy alternative that provided 10,000 L/month of water for free unconditionally was available as an option in the block.	1 0	Available as an option in the block Not available as an option in the block
CurrentPolicy	All entries had a value of 1 as the current policy option was always available.	1	Always available